

Online Car Rental Database.

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IFSC 33003-01 - Curr Trends in Database Tech

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Introduction

The Online Car Rental Database is a project designed using Oracle and Lucidchart. In this project, we were assigned to design and develop a database for an online car rental system. We began by creating a detailed description of the database, including its attributes, and then transformed this into an Entity-Relationship (ER) diagram. From the ER diagram, we translated the design into a relational schema and analyzed it to ensure it met Third Normal Form (3NF). Finally, we implemented the database by creating tables in Oracle and developed eight queries to demonstrate its functionality.

Description

When describing the sample database, I had to consider what is typically included in car rental systems. Common entities in rental databases include customer, car, rental, location, and payment. Each entity contains specific attributes that are essential for managing the system. Through research, I identified which details are commonly required when renting a car and used that to define my data requirements.

Entities and Data Requirements

Customer Information:

- CustomerID
- Name
- PhoneNumber
- Email
- DriverLicenseNum

Car:

- CarID
- Make
- Model
- Year
- Type
- RentalPricePerDay

Rental:

- RentalID
- CustomerID
- PickupLocationID

- PickUpDate
- ReturnLocationID
- ReturnDate
- TotalPayment

Location:

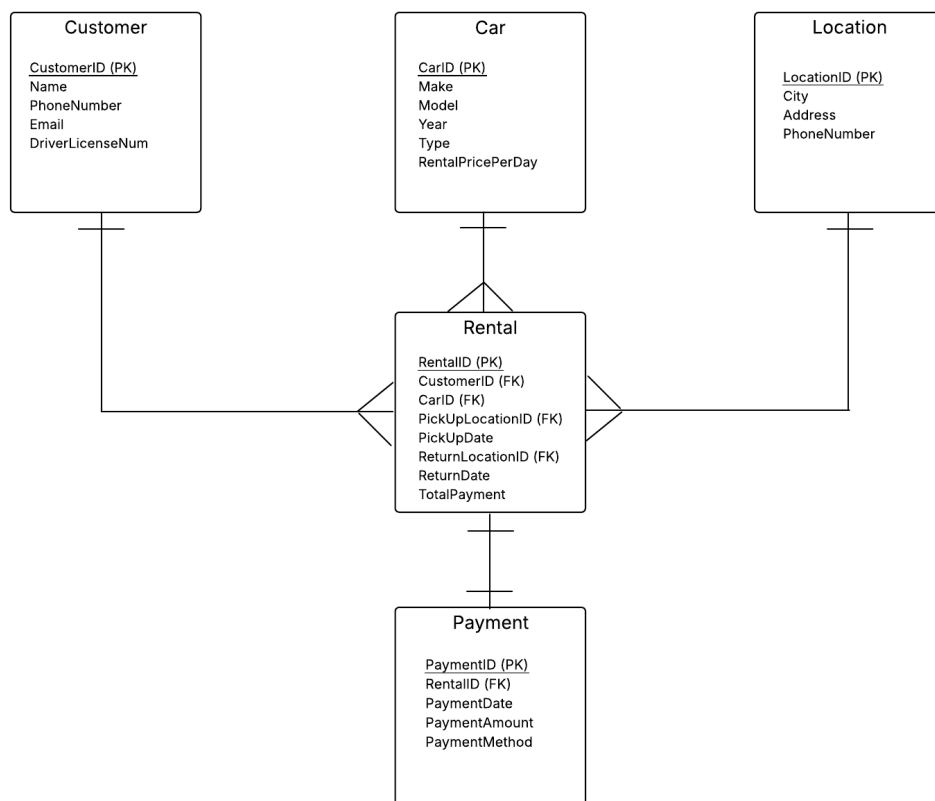
- LocationID
- City
- Address
- PhoneNumber

Payment:

- PaymentID
- RentalID
- PaymentDate
- PaymentAmount
- PaymentMethod

Entity-Relationship (ER) diagram

When creating the ER diagram, I had to identify the primary keys for each entity and determine the cardinality of the relationships between them. Using Lucidchart, I was able to design and visualize the ER diagram effectively.



Relational Schema

Next, I translated the ER diagram into a relational schema using Lucidchart, converting each entity and relationship into tables with appropriate primary and foreign keys.

Customer(CustomerID , Name, PhoneNumber, Email, DriverLicenseNum)

Car(CarID, Make, Model, Year, Type, RentalPricePerDay)

Location(LocationID, City, Address, PhoneNumber)

Rental (RentalID, CustomerID (FK), CarID(FK), PickUpLocationID(FK),
ReturnLocationID(FK), PickUpDate, ReturnDate, TotalPayment)

Payment (PaymentID, RentalID(FK), PaymentDate, PaymentAmount, PaymentMethod)

First Normal Form (1NF):

The database satisfies 1NF because all attributes are atomic, and there are no multi-valued or repeating groups. Each column stores a single value, which keeps the data organized. For example, in the Car table, attributes such as Model, Year, and Type are stored in separate columns rather than combined into one. This ensures that each field contains only one value and improves clarity and structure in the database.

Second Normal Form (2NF):

To check if the database is in 2NF, it must first satisfy 1NF. Since the database already meets 1NF, we then check for partial dependencies. A database is in 2NF if all non-key attributes are fully dependent on the primary key. In this database, each table uses a single-column primary key. For example, in the Car table, attributes such as Make, Model, and RentalPricePerDay all depend on CarID. Since there are no composite keys, there are no partial dependencies, so the database satisfies 2NF.

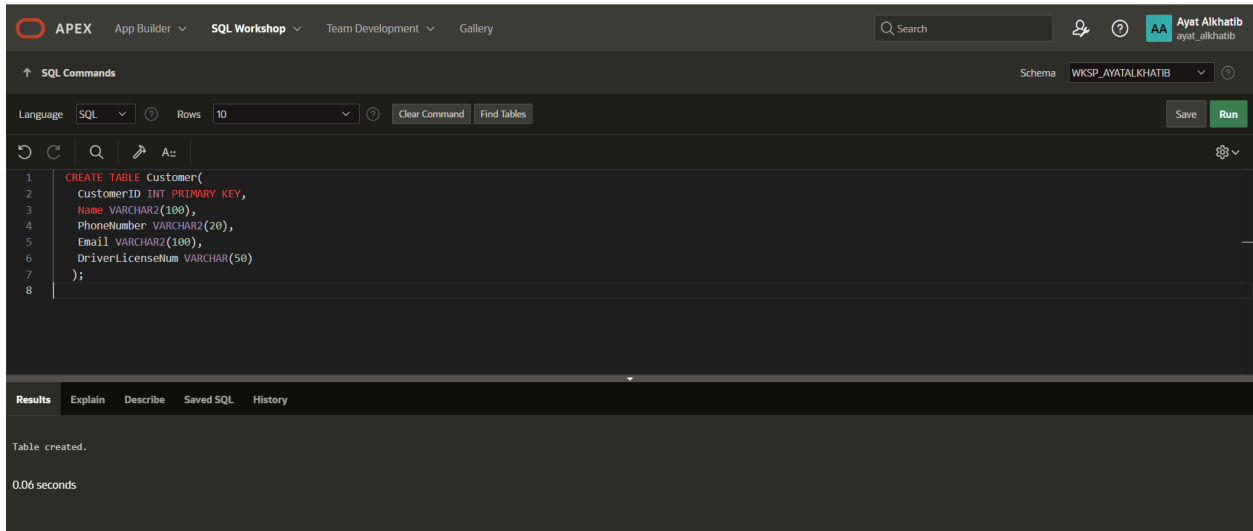
Third Normal Form (3NF):

To check if the database is in 3NF, it must satisfy both 1NF and 2NF. Since the database meets these requirements, we then check for transitive dependencies. A database is in 3NF if no non-primary key attribute depends on another non-primary key attribute. In this database, all non-key attributes depend only on their primary key, meaning there are no transitive dependencies. For instance, in the Location table, the Address and PhoneNumber are determined strictly by the LocationID. Knowing the Address does not automatically determine the PhoneNumber (and vice versa). Both attributes depend only on the primary key, which avoids any transitive dependency.

Implementation of the databases

For the implementation of the database, I chose Oracle, as it is the primary environment used in this course and supports relational constraints, data integrity, and complex queries. Using Data Definition Language (DDL), I translated the relational schema into physical tables. I then populated the five tables with six tuples each.

Customer:

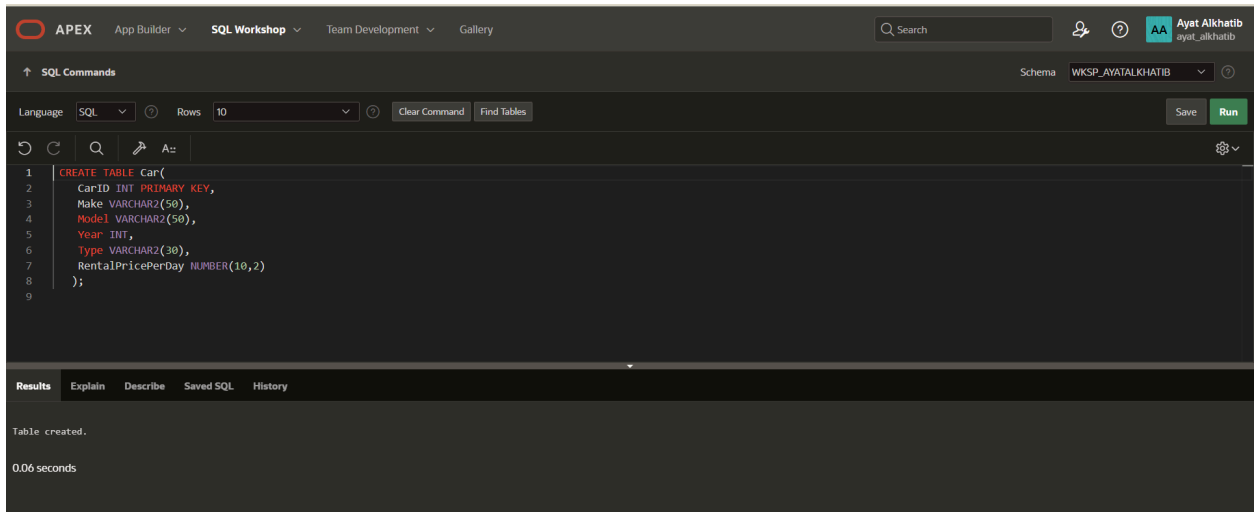


The screenshot displays the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Ayat Alkhatib' are on the right. The 'SQL Commands' section is active, showing a schema dropdown set to 'WKSP_AVATALKHATIB'. Below this, the 'Language' is set to 'SQL' and 'Rows' to '10'. The main editor contains the following SQL command:

```
1 CREATE TABLE Customer(  
2   CustomerID INT PRIMARY KEY,  
3   Name VARCHAR2(100),  
4   PhoneNumber VARCHAR2(20),  
5   Email VARCHAR2(100),  
6   DriverLicenseNum VARCHAR(50)  
7 );  
8
```

The bottom section, labeled 'Results', shows the message 'Table created.' and a execution time of '0.06 seconds'. Navigation tabs for 'Results', 'Explain', 'Describe', 'Saved SQL', and 'History' are visible.

Car:

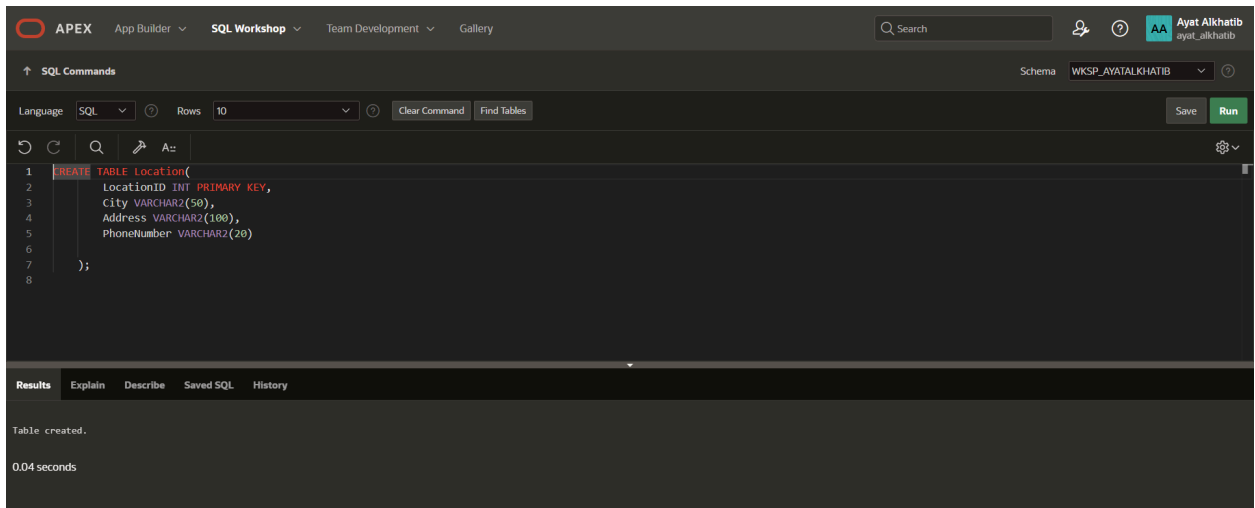


The screenshot shows the APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Ayat Alkhatib' are on the right. The 'SQL Commands' tab is active, showing a schema dropdown set to 'WKSP_AVATALKHATIB'. The language is 'SQL' and the number of rows is '10'. The SQL command area contains the following code:

```
1 CREATE TABLE Car(  
2   CarID INT PRIMARY KEY,  
3   Make VARCHAR2(50),  
4   Model VARCHAR2(50),  
5   Year INT,  
6   Type VARCHAR2(30),  
7   RentalPricePerDay NUMBER(10,2)  
8 );  
9
```

The 'Results' tab is selected, displaying the message 'Table created.' and the execution time '0.06 seconds'.

Location:



The screenshot shows the APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Ayat Alkhatib' are on the right. The 'SQL Commands' tab is active, showing a schema dropdown set to 'WKSP_AVATALKHATIB'. The language is 'SQL' and the number of rows is '10'. The SQL command area contains the following code:

```
1 CREATE TABLE Location(  
2   LocationID INT PRIMARY KEY,  
3   City VARCHAR2(50),  
4   Address VARCHAR2(100),  
5   PhoneNumber VARCHAR2(20)  
6 );  
7  
8
```

The 'Results' tab is selected, displaying the message 'Table created.' and the execution time '0.04 seconds'.

Rental:

The screenshot shows the APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Ayat Alkhatib' are on the right. The 'SQL Commands' panel is active, showing a schema dropdown set to 'WKSP_AYATALKHATIB'. The command area contains the following SQL code:

```
1 CREATE TABLE Rental(  
2   RentalID INT PRIMARY KEY,  
3   CustomerID INT,  
4   CarID INT,  
5   PickupLocationID INT,  
6   ReturnLocationID INT,  
7   PickupDate DATE,  
8   ReturnDate DATE,  
9   TotalPayment NUMBER(10,2),  
10  CONSTRAINT fk_cust FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),  
11  CONSTRAINT fk_car FOREIGN KEY (CarID) REFERENCES Car(CarID),  
12  CONSTRAINT fk_pickup_loc FOREIGN KEY (PickupLocationID) REFERENCES Location(LocationID),  
13  CONSTRAINT fk_return_loc FOREIGN KEY (ReturnLocationID) REFERENCES Location(LocationID)  
14 );  
15  
16  
17
```

The 'Results' panel at the bottom shows the message 'Table created.' and the execution time '0.08 seconds'.

Payment:

The screenshot shows the APEX SQL Workshop interface. The top navigation bar is the same as the previous screenshot. The 'SQL Commands' panel is active, showing a schema dropdown set to 'WKSP_AYATALKHATIB'. The command area contains the following SQL code:

```
1 CREATE TABLE Payment(  
2   PaymentID INT PRIMARY KEY,  
3   RentalID INT,  
4   PaymentDate INT,  
5   PaymentAmount NUMBER(10,2),  
6   PaymentMethod VARCHAR2(30),  
7   CONSTRAINT fk_rental_pay FOREIGN KEY (RentalID) REFERENCES Rental(RentalID)  
8 );  
9  
10  
11
```

The 'Results' panel at the bottom shows the message 'Table created.' and the execution time '0.08 seconds'.

After implementing the tables, the next step was to populate them by inserting values into each table.

Customers:

The Customers table includes the following attributes: Name, PhoneNumber, Email, and DriverLicenseNum. The data inserted into the table is as follows:

- Olivia Brown, 501-111-1111, oliviabrown@gmail.com, AR-123456
- Noah Jones, 501-222-2222, noahjones@gmail.com, AR-234567
- Liam Smith, 501-333-3333, liamsmith@gmail.com, AR-345678
- James Williams, 479-444-4444, jameswilliams@gmail.com, AR-456789
- Asher Johnson, 479-555-5555, asherjohnson@gmail.com, AR-567891
- Charlotte Davis, 501-666-6666, charlottedavis@gmail.com, AR-678912

The screenshot displays the APEX SQL Workshop interface. At the top, the navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Ayat Alkhatib' are on the right. The main area is titled 'SQL Commands' and shows a schema 'WKSP_AYATALKHATIB'. The command editor contains the following SQL code:

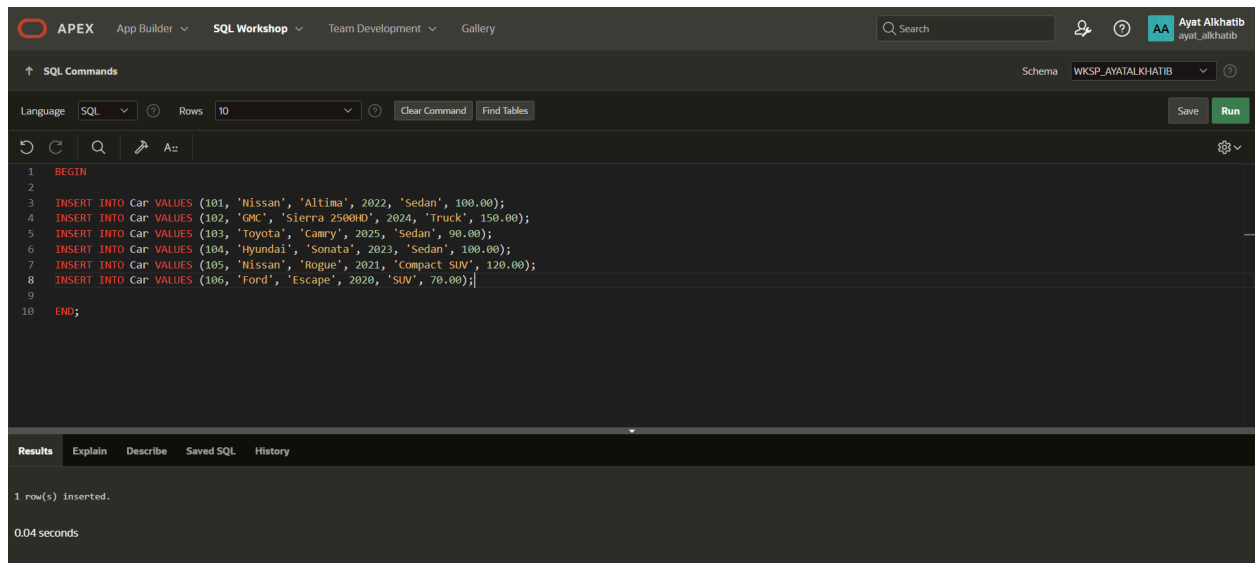
```
1 BEGIN
2
3 INSERT INTO Customer VALUES (1, 'Olivia Brown', '501-111-1111', 'oliviabrown@gmail.com', 'AR-123456');
4 INSERT INTO Customer VALUES (2, 'Noah Jones', '501-222-2222', 'noahjones@gmail.com', 'AR-234567');
5 INSERT INTO Customer VALUES (3, 'Liam Smith', '501-333-3333', 'liamsmith@gmail.com', 'AR-345678');
6 INSERT INTO Customer VALUES (4, 'James Williams', '479-444-4444', 'jameswilliams@gmail.com', 'AR-456789');
7 INSERT INTO Customer VALUES (5, 'Asher Johnson', '479-555-5555', 'asherjohnson@gmail.com', 'AR-567891');
8 INSERT INTO Customer VALUES (6, 'Charlotte Davis', '501-666-6666', 'charlottedavis@gmail.com', 'AR-678912');
9
10 END;
```

Below the editor, the 'Results' tab is active, showing '1 row(s) inserted.' and a duration of '0.01 seconds'. Other tabs like 'Explain', 'Describe', 'Saved SQL', and 'History' are also visible.

Car:

The Car table includes the following attributes: CarID, Make, Model, Year, Type, and RentalPricePerDay. The data inserted into the table is as follows:

- 101, Nissan, Altima, 2022, Sedan, 100.00
- 102, GMC, Sierra 2500HD, 2024, Truck, 150.00
- 103, Toyota, Camry, 2025, Sedan, 90.00
- 104, Hyundai, Sonata, 2023, Sedan, 100.00
- 105, Nissan, Rogue, 2021, Compact SUV, 120.00
- 106, Ford, Escape, 2020, SUV, 70.00



The screenshot displays the APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Ayat Alkhatib' are on the right. The 'SQL Commands' panel shows a schema dropdown set to 'WKSP_AYATALKHATIB'. Below this, the 'Language' is set to 'SQL' and 'Rows' to '10'. The SQL command window contains the following code:

```
1 BEGIN
2
3 INSERT INTO Car VALUES (101, 'Nissan', 'Altima', 2022, 'Sedan', 100.00);
4 INSERT INTO Car VALUES (102, 'GMC', 'Sierra 2500HD', 2024, 'Truck', 150.00);
5 INSERT INTO Car VALUES (103, 'Toyota', 'Camry', 2025, 'Sedan', 90.00);
6 INSERT INTO Car VALUES (104, 'Hyundai', 'Sonata', 2023, 'Sedan', 100.00);
7 INSERT INTO Car VALUES (105, 'Nissan', 'Rogue', 2021, 'Compact SUV', 120.00);
8 INSERT INTO Car VALUES (106, 'Ford', 'Escape', 2020, 'SUV', 70.00);
9
10 END;
```

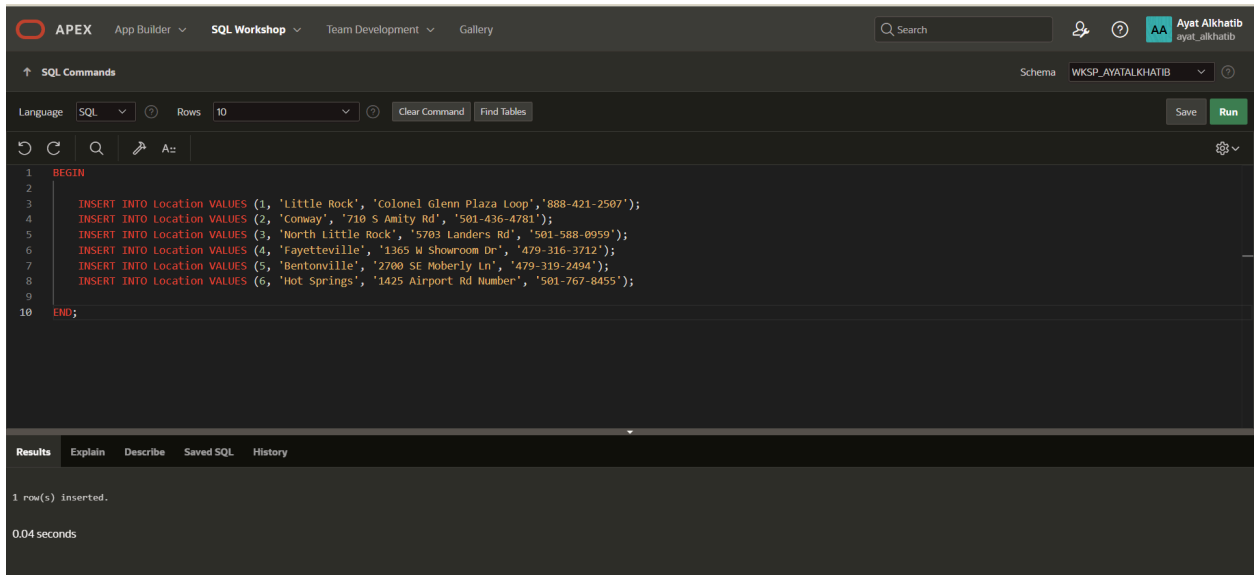
The 'Results' panel at the bottom shows the execution output: '1 row(s) inserted.' and '0.04 seconds'.

Location:

The Location table includes the following attributes: City, Address, and PhoneNumber.

The chosen locations for the database are as follows:

- Little Rock, 401 Colonel Glenn Plaza Loop, 888-421-2507
- Conway, 710 S Amity Rd, 501-436-4781
- North Little Rock, 5703 Landers Rd, 501-588-0959
- Fayetteville, 1365 W Showroom Dr, 479-316-3712
- Bentonville, 2700 SE Moberly Ln, 479-319-2494
- Hot Springs, 1425 Airport Rd, 501-767-8455



The screenshot shows the APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Ayat Alkhatib' are on the right. The 'SQL Commands' section is active, showing a schema dropdown set to 'WKSP_AYATALKHATIB'. The SQL editor contains the following code:

```
1 BEGIN
2
3   INSERT INTO Location VALUES (1, 'Little Rock', 'Colonel Glenn Plaza Loop', '888-421-2507');
4   INSERT INTO Location VALUES (2, 'Conway', '710 S Amity Rd', '501-436-4781');
5   INSERT INTO Location VALUES (3, 'North Little Rock', '5703 Landers Rd', '501-588-0959');
6   INSERT INTO Location VALUES (4, 'Fayetteville', '1365 W Showroom Dr', '479-316-3712');
7   INSERT INTO Location VALUES (5, 'Bentonville', '2700 SE Moberly Ln', '479-319-2494');
8   INSERT INTO Location VALUES (6, 'Hot Springs', '1425 Airport Rd Number', '501-767-8455');
9
10 END;
```

Below the editor, the 'Results' tab is selected, displaying the output: '1 row(s) inserted.' and '0.04 seconds'.

Rental:

The Rental table includes RentalID, CustomerID, CarID, PickupLocationID, ReturnLocationID, PickupDate, ReturnDate, and TotalPayment. It records each customer's rental transaction, including the car rented, locations, rental period, and total cost.

- Olivia Brown (RentalID 1001) rented Car 101 from April 20–25, 2026, for 500.00, with pickup and return at Location 1.
- Noah Jones (1002) rented Car 102 from April 22–24, 2026, for 300.00, with pickup at Location 2 and return at Location 3.
- Liam Smith (1003) rented Car 103 from April 25–28, 2026, for 270.00, with both pickup and return at Location 3.
- James Williams (1004) rented Car 104 from April 28 to May 1, 2026, for 300.00, with pickup at Location 4 and return at Location 5.
- Asher Johnson (1005) rented Car 105 from May 1–3, 2026, for 240.00, with both pickup and return at Location 5.
- Charlotte Davis (1006) rented Car 106 from May 1–4, 2026, for 210.00, with pickup and return at Location 6.

The screenshot displays the Oracle APEX SQL Workshop interface. The top navigation bar includes the APEX logo and tabs for App Builder, SQL Workshop (selected), Team Development, and Gallery. A search bar is located on the right. Below the navigation bar, the 'SQL Commands' section is active, showing the schema 'WKSP_AYATALKHATIB'. The 'Language' is set to 'SQL' and 'Rows' is set to '10'. The 'Run' button is highlighted in green. The SQL editor contains the following code:

```
1 BEGIN
2
3 INSERT INTO Rental VALUES (1001, 1, 101, 1, 1, TO_DATE('2026-04-20', 'YYYY-MM-DD'), TO_DATE('2026-04-25', 'YYYY-MM-DD'), 500.00);
4 INSERT INTO Rental VALUES (1002, 2, 102, 2, 3, TO_DATE('2026-04-22', 'YYYY-MM-DD'), TO_DATE('2026-04-24', 'YYYY-MM-DD'), 300.00);
5 INSERT INTO Rental VALUES (1003, 3, 103, 3, 3, TO_DATE('2026-04-25', 'YYYY-MM-DD'), TO_DATE('2026-04-28', 'YYYY-MM-DD'), 270.00);
6 INSERT INTO Rental VALUES (1004, 4, 104, 4, 5, TO_DATE('2026-04-28', 'YYYY-MM-DD'), TO_DATE('2026-05-01', 'YYYY-MM-DD'), 300.00);
7 INSERT INTO Rental VALUES (1005, 5, 105, 5, 5, TO_DATE('2026-05-01', 'YYYY-MM-DD'), TO_DATE('2026-05-03', 'YYYY-MM-DD'), 240.00);
8 INSERT INTO Rental VALUES (1006, 6, 106, 6, 6, TO_DATE('2026-05-01', 'YYYY-MM-DD'), TO_DATE('2026-05-04', 'YYYY-MM-DD'), 210.00);
9
10 END;
```

The 'Results' tab is selected, showing the output: '1 row(s) inserted.' and '0.01 seconds'.

Payment:

The Payment table includes PaymentID, RentalID, PaymentDate, PaymentAmount, and PaymentMethod. The system supports two payment types: credit cards and debit cards.

- Olivia Brown (PaymentID 501) paid 500.00 on 2026-04-25 using a Credit Card.
- Noah Jones (502) paid 300.00 on 2026-04-24 using a Debit Card.
- Liam Smith (503) paid 270.00 on 2026-04-28 using a Debit Card.
- James Williams (504) paid 300.00 on 2026-05-01 using a Credit Card.
- Asher Johnson (505) paid 240.00 on 2026-05-02 using a Credit Card.
- Charlotte Davis (506) paid 210.00 on 2026-05-02 using a Credit Card.

The screenshot displays the APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Ayat Alkhatib' are on the right. The 'SQL Commands' panel shows a schema dropdown set to 'WKSP_AYATALKHATIB'. Below this, the 'Language' is set to 'SQL' and 'Rows' to '10'. The main SQL editor contains the following code:

```

1 BEGIN
2
3 INSERT INTO Payment (PaymentID, RentalID, PaymentDate, PaymentAmount, PaymentMethod) VALUES (501, 1001, 20260425, 500.00, 'Credit Card');
4 INSERT INTO Payment (PaymentID, RentalID, PaymentDate, PaymentAmount, PaymentMethod) VALUES (502, 1002, 20260424, 300.00, 'Debit Card');
5 INSERT INTO Payment (PaymentID, RentalID, PaymentDate, PaymentAmount, PaymentMethod) VALUES (503, 1003, 20260428, 270.00, 'Debit Card');
6 INSERT INTO Payment (PaymentID, RentalID, PaymentDate, PaymentAmount, PaymentMethod) VALUES (504, 1004, 20260501, 300.00, 'Credit Card');
7 INSERT INTO Payment (PaymentID, RentalID, PaymentDate, PaymentAmount, PaymentMethod) VALUES (505, 1005, 20260502, 240.00, 'Credit Card');
8 INSERT INTO Payment (PaymentID, RentalID, PaymentDate, PaymentAmount, PaymentMethod) VALUES (506, 1006, 20260502, 210.00, 'Credit Card');
9
10 END;

```

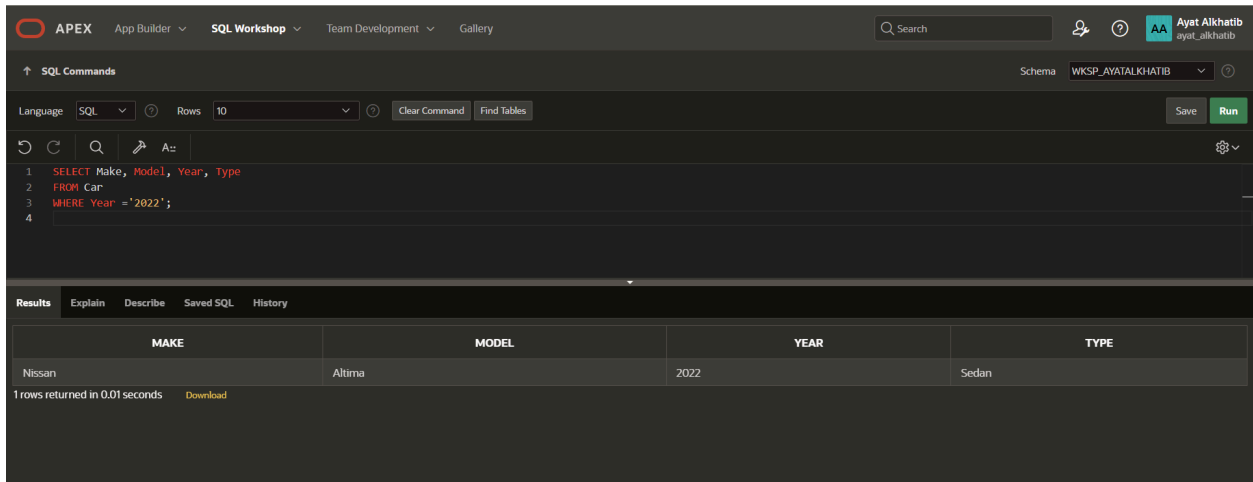
The bottom 'Results' panel shows the execution outcome: '1 row(s) inserted.' and a duration of '0.04 seconds'. Navigation tabs for 'Results', 'Explain', 'Describe', 'Saved SQL', and 'History' are visible above the results.

SQL queries

I created 8 SQL queries to demonstrate different functionalities of the database, including retrieving customer, car, rental, location, and payment information, as well as performing joins and aggregations for analysis.

1. Cars by Year:

Finds cars from a specific year.



The screenshot shows the APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Ayat Alkhatib' are on the right. The 'SQL Commands' section shows a query:

```
1 SELECT Make, Model, Year, Type
2 FROM Car
3 WHERE Year = '2022';
4
```

 The 'Results' tab is active, displaying a table with one row:

MAKE	MODEL	YEAR	TYPE
Nissan	Altima	2022	Sedan

 Below the table, it says '1 rows returned in 0.01 seconds' and has a 'Download' link.

2. Rentals from Little Rock:

Finds all locations in Little Rock.

The screenshot shows the APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Ayat Alkhatib' are on the right. The 'SQL Commands' tab is active, showing a query:

```
1 SELECT LocationID, City, Address, PhoneNumber
2 FROM Location
3 where City = 'Little Rock';
4
5
6
```

 The 'Run' button is highlighted. Below the query, the 'Results' tab shows a table with 1 row:

LOCATIONID	CITY	ADDRESS	PHONENUMBER
1	Little Rock	Colonel Glenn Plaza Loop	888-421-2507

 The status bar indicates '1 rows returned in 0.02 seconds'.

3. Debit Card Users:

Displays customers who paid using a debit card.

The screenshot shows the APEX SQL Workshop interface. The top navigation bar is the same as the previous screenshot. The 'SQL Commands' tab is active, showing a query:

```
1 SELECT
2     c.Name,
3     p.PaymentMethod
4 FROM Payment p
5 JOIN Rental r ON p.RentalID = r.RentalID
6 JOIN Customer c ON r.CustomerID = c.CustomerID
7 where p.PaymentMethod = 'Debit Card';
8
9
10
```

 The 'Run' button is highlighted. Below the query, the 'Results' tab shows a table with 2 rows:

NAME	PAYMENTMETHOD
Noah Jones	Debit Card
Liam Smith	Debit Card

 The status bar indicates '2 rows returned in 0.04 seconds'.

4. Customers with 501 Phone Numbers:

Filters customers based on phone number prefix.

The screenshot shows the APEX SQL Workshop interface. The SQL command is as follows:

```

1 SELECT CustomerID, Name, PhoneNumber, Email, DriverLicenseNum
2 FROM Customer
3 WHERE PhoneNumber LIKE '501%';
4
5
6
7

```

The results tab shows 4 rows returned in 0.01 seconds. The data is as follows:

CUSTOMERID	NAME	PHONENUMBER	EMAIL	DRIVERLICENSENUM
2	Noah Jones	501-222-2222	noahjones@gmail.com	AR-234567
3	Liam Smith	501-333-3333	liamsmith@gmail.com	AR-345678
6	Charlotte Davis	501-666-6666	charlottedavis@gmail.com	AR-678912
1	Olivia Brown	501-111-1111	oliviabrown@gmail.com	AR-123456

5. Full Rental Details:

Combines customer, car, and pickup location information.

The screenshot shows the APEX SQL Workshop interface. The SQL command is as follows:

```

1
2 SELECT
3   c.Name AS Customer_Name,
4   car.Make,
5   car.Model,
6   l.City AS Pickup_City,
7   r.PickupDate
8 FROM Rental r
9 JOIN Customer c ON r.CustomerID = c.CustomerID
10 JOIN Car car ON r.CarID = car.CarID
11 JOIN Location l ON r.PickupLocationID = l.LocationID;
12

```

The results tab shows 6 rows returned in 0.03 seconds. The data is as follows:

CUSTOMER_NAME	MAKE	MODEL	PICKUP_CITY	PICKUPDATE
Olivia Brown	Nissan	Altima	Little Rock	4/20/2026
Noah Jones	GMC	Sierra 2500HD	Conway	4/22/2026
Liam Smith	Toyota	Camry	North Little Rock	4/25/2026
James Williams	Hyundai	Sonata	Fayetteville	4/28/2026
Asher Johnson	Nissan	Rogue	Bentonville	5/1/2026
Charlotte Davis	Ford	Escape	Hot Springs	5/1/2026

6. Payment Summary:

Shows total earnings and number of payments by method.

The screenshot shows the APEX SQL Workshop interface. The SQL command is as follows:

```

1 SELECT
2   PaymentMethod,
3   SUM(PaymentAmount) AS Total_Earning,
4   COUNT (*) AS Number_of_payments
5 FROM Payment
6 GROUP BY PaymentMethod;

```

The results are displayed in a table with the following data:

PAYMENTMETHOD	TOTAL_EARNING	NUMBER_OF_PAYMENTS
Credit Card	1250	4
Debit Card	570	2

2 rows returned in 0.01 seconds [Download](#)

7. High Value Customers:

Finds customers who spent more than 250 on a rental.

The screenshot shows the APEX SQL Workshop interface. The SQL command is as follows:

```

1 SELECT
2   c.Name,
3   r.TotalPayment,
4   r.PickUpDate
5 FROM Rental r
6 JOIN Customer c ON r.CustomerID = c.CustomerID
7 WHERE r.TotalPayment > 250;

```

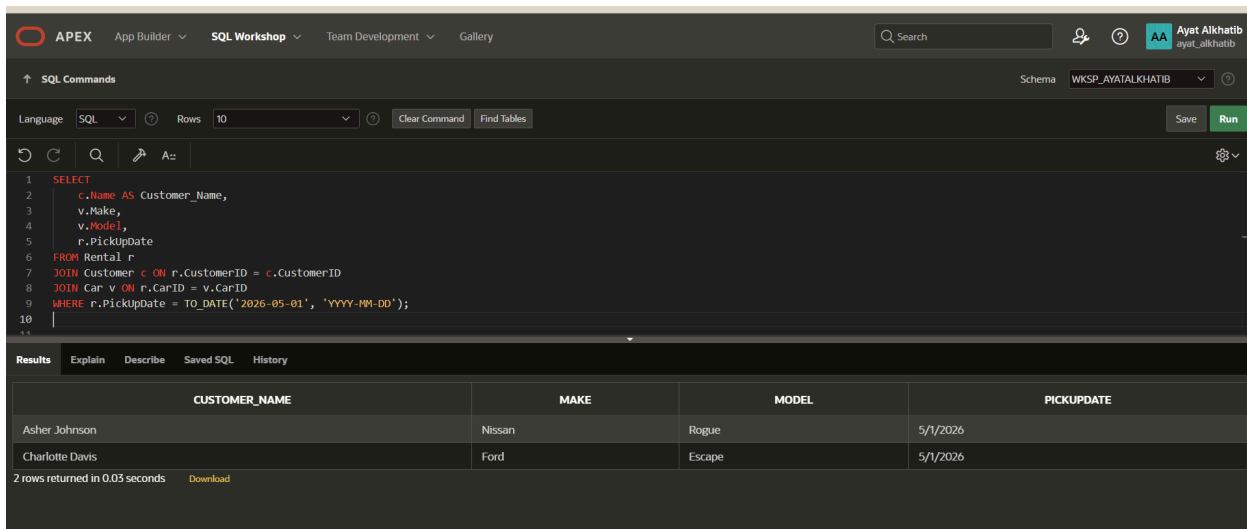
The results are displayed in a table with the following data:

NAME	TOTALPAYMENT	PICKUPDATE
Noah Jones	300	4/22/2026
Liam Smith	270	4/25/2026
James Williams	300	4/28/2026
Olivia Brown	500	4/20/2026

4 rows returned in 0.02 seconds [Download](#)

8. Rentals on May 1st:

Shows customers renting on a specific date.



The screenshot displays the APEX SQL Workshop interface. At the top, there's a navigation bar with 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Ayat Alkhatib' are on the right. Below this, the 'SQL Commands' section shows a query editor with the following SQL code:

```
1 SELECT
2   c.Name AS Customer_Name,
3   v.Make,
4   v.Model,
5   r.PickupDate
6 FROM Rental r
7 JOIN Customer c ON r.CustomerID = c.CustomerID
8 JOIN Car v ON r.CarID = v.CarID
9 WHERE r.PickupDate = TO_DATE('2026-05-01', 'YYYY-MM-DD');
```

Below the editor, the 'Results' tab is active, showing a table with 2 rows. The table has columns: CUSTOMER_NAME, MAKE, MODEL, and PICKUPDATE. The data is as follows:

CUSTOMER_NAME	MAKE	MODEL	PICKUPDATE
Asher Johnson	Nissan	Rogue	5/1/2026
Charlotte Davis	Ford	Escape	5/1/2026

At the bottom of the results section, it states '2 rows returned in 0.03 seconds' and provides a 'Download' link.

Conclusion

To conclude, the database functions effectively, with SQL queries successfully displaying the inserted values. The ER diagram clearly represents the structure and relationships within the system, while the relational schema helped translate the design into functional SQL code. Overall, the project demonstrates a complete database development process from design to implementation.